

Geometric model for the integral cohomology of toric hypersurface arrangements

Peter Xu

Abstract

We give a self-contained proof of a geometric presentation for the integral cohomology ring of an arbitrary toric hypersurface complement, in terms of elementary conical chamber data. We describe explicit Borel–Moore cycle and log-differential form representatives; the latter gives a new, independent proof of formality over $\mathbb{Q}$ of such arrangements, as well as torsion-freeness over $\mathbb{Z}$.

Contents

1	Introduction	1
1.1	Connections in the literature	2
1.2	Acknowledgements	3
2	Setup and notation	3
3	The cocharacter presentation	4
3.1	Geometric realization	5
4	Proof of the presentation	6
4.1	Chambers and Orlik–Solomon classes	6
4.2	The Leray spectral sequence	7
4.3	Degeneration and the proof of the integral presentation	8
5	The character formulation and de Rham realization	9
5.1	Admissible cones and their decorations	9
5.2	Conical generating functions	9
5.3	Differential form representatives	11
5.4	Multiplicative structure	12

1 Introduction

Some very pretty classical results capture finely the topological structure of complex *hyperplane complements*, i.e. the complement of a union of hyperplanes H inside affine (or projective, in a difference only of formulation) $\mathbb{C}^n$. In particular, the cohomology ring of $\mathbb{C}^n - H$ is formal and well-understood (including its integral structure via Betti periods) via the Orlik–Solomon model [OS], depending on only the underlying combinatorics of independence between linear forms (the *matroid* data).

Replacing affine space with $\mathbf{T}$, a split algebraic torus, i.e. a power of the multiplicative group, we can ask the same about the complement of a union H of hypersurfaces coming from codimension-1 tori embedded affinely (i.e. as translates of images of homomorphisms); we will refer to these as “toric hyperplanes”. More generally, we will refer to “toric subspaces” as affinely embedded tori without condition on codimension.

In the toric setting, there is the new feature of *non-unimodularity*, that intersections of toric hypersurfaces may be disconnected unions of several toric subspaces.

To give a very condensed summary of work in this toric setting, De Concini–Procesi [DP] gave a clean description of the cohomology of $\mathbf{T} - H$ in the case of a unimodular arrangement without this phenomenon, reminiscent of the Orlik–Solomon model. Callegaro et al. [CDDMP] (with a small corrigendum in [CDDMP2]) generalized this description to the general setting, giving an integral presentation of the full cohomology ring, via a formality isomorphism with explicitly defined differential forms.

Inspired by our previous work giving a combinatorial/geometric symbolic formalism on certain special toric arrangement complements [X1], [RX], which are closely related to the arithmetic of “Eisenstein cocycles,” we give an analogous parameterization of the cohomology of a toric hyperplane complement, independent of and equivalent to the presentation constructed in [CDDMP].

Our formalism has many nice features:

- We find the conical geometry greatly simplifies and clarifies the structure of the presentation compared to [CDDMP], whose heavy notation involves considerable amounts of auxiliary bookkeeping, some of which is avoidable.
- Our geometric formalism parameterizes not only differential forms, but special choices of Poincaré dual cycles in corresponding relative homology groups; this may be especially relevant in prospective work on Aomoto polylogarithms coming from the integration pairing of these against differential forms.
- The geometric formalism naturally sees important structure on the cohomology ring: in particular, the filtration by lineality (dimension of contained subspace) on conical chambers exactly is mapped to the Leray filtration, and the graded pieces recover the description of local cohomology in terms of Orlik–Solomon complexes.
- As a result of this geometric picture, we are able to not only independently prove the validity of our presentation, but the formality and torsion-freeness of cohomology (the latter of which is used as an input in [CDDMP]).¹

Our exposition in fact *first* constructs and proves the isomorphism (Theorem 3.2) using explicit cycles in homology, before turning to the realization as differential forms in Section 5.

1.1 Connections in the literature

We believe the conical chamber formalism can also be useful in exploring and computing aspects of Brown–Dupont’s “genus zero positive geometry” [BD] for the toric arrangements. Our own primary interest is to use our parameterization to understand certain period pairings between our

¹A previous version of this article described our results as “formality over $\mathbb{Z}$,” language also used in the abstract of loc. cit. However, formality over $\mathbb{Z}$ in the sense of a dg-algebra isomorphism between an integral cochain complex and the integral cohomology fails even already for many ordinary hyperplane arrangements; what we meant (and was presumably meant in [CDDMP]) instead was simply the combination of formality over $\mathbb{Q}$ and a presentation recovering the integral structure. Thanks to Clément Dupont for pointing this out.

combinatorially-indexed forms and cycles on subspaces of $\mathbf{T}$ yielding the “toric Aomoto polylogarithms” whose relevance to mixed Tate motives over cyclotomic fields is recently becoming an object of interest [CRR] [X2].

We remark also that, extending the Milnor K -theory calculation of [X2], we expect that a similar presentation can even be written down for the *motivic* cohomology of a torus arrangement complement over a field F , using induction along residue maps in a coniveau spectral sequence (instead of the Leray sequences). In this case, the presentation should look very similar to the present one, but with bigraded coefficient ring $H^\bullet(F, \mathbb{Z}(\bullet))$ —or, rather, fitting better with our language, over the torus cohomology $H^\bullet(\mathbf{T}_F, \mathbb{Z}(\bullet))$ (which is “almost” an exterior algebra over the former up to 2-torsion phenomena by the fundamental theorem in motivic cohomology/ K -theory). We expect the only real difference to be in the multiplicative structure, due to the phenomenon that $\{z, z\}$ is nonzero (unlike $d \log z \wedge d \log z$) and $\{1 - z, c - z\}$ is not even torsion for a constant $c \neq 0, 1$ (unlike $d \log(1 - z) \wedge d \log(c - z)$).

1.2 Acknowledgements

Much thanks to Clément Dupont, Anaëlle Pfister, and Claudia Yun for interesting conversations, and Roberto Pagaria for replying to an email.

In accordance with the Leiden Declaration: thanks also to Claude Opus and ChatGPT Astra, which were very useful conversation partners and helped correct several formulas and conventions. In particular, a significant conceptual contribution was to suggest bypassing existing torsion-freeness or integrality results (as cited in [CDDMP]) by working with Borel–Moore cycles, as the Leray spectral sequence/filtration argument I originally wrote in the de Rham cohomology setting can be run just as easily with $\mathbb{Z}$ -coefficients, which I did not realize originally.

2 Setup and notation

Let Λ be a rank- n lattice, $\mathbf{T} = \text{Spec } \mathbb{C}[\Lambda]$ our algebraic torus with character lattice Λ , and write χ^v for the character corresponding to $v \in \Lambda$. Fix a finite set E (which, for later comparison purposes, we endow with an ordering), which will parameterize our toric hyperplanes; these are indexed by a collection of character vectors u_i and translation scalars $b_i \in \mathbb{C}^\times$, both for $i \in E$. Then we set the notation, chosen in part to align with [CDDMP],

$$H_i = \{\chi^{u_i} = b_i\}, \quad \mathcal{A} = \{H_i, i \in E\}, \quad H = \bigcup_{i \in E} H_i, \quad M = M(\mathcal{A}) := \mathbf{T} - H,$$

and

$$\psi_v = \frac{1}{2\pi\sqrt{-1}} d \log \chi^v, \quad R = H^\bullet(\mathbf{T}; \mathbb{Z}) = \bigwedge^\bullet \Lambda.$$

We will, a bit abusively, reuse the same notation for an exterior tensor power and the corresponding differential form.

Let $\mathcal{W}$ (the set of “layers”) be the set of connected components of intersections of hypertori, including the whole space $\mathbf{T}$, ordered by reverse inclusion. For $W \in \mathcal{W}$, set $\Lambda_W \subset \Lambda$ to be the saturated rank-codim W sublattice of characters constant along W .

For a vector v in this sublattice, we set $\lambda_W(v) = \chi^v(p)^{-1}$ for any choice of $p \in W$; this is a character of Λ_W . Write S_W for the subset of E corresponding to hyperplanes containing W .

We may see that Λ_W is the saturation of the span of u_i for $i \in S_W$, that $\lambda_W(u_i) = b_i^{-1}$ for each $i \in S_W$, and we have a canonical identification

$$H^\bullet(W; \mathbb{Z}) = \bigwedge^\bullet (\Lambda / \Lambda_W).$$

For $A \subset E$, set

$$\Lambda_A^0 = \langle u_i : i \in A \rangle,$$

Λ_A for its saturation, and $m(A) = [\Lambda_A : \Lambda_A^0]$; this is a measure of the non-unimodularity of the subarrangement A .

Note that the intersection indexed by A may be empty; it is nonempty exactly when we have the coherence condition for any integer coefficients $a_i, i \in A$

$$\sum_{i \in A} a_i u_i = 0 \Rightarrow \prod_{i \in A} b_i^{a_i} = 1 \quad (1)$$

The set of components, in this case, then is in natural bijection with the extensions of the assignment $u_i \mapsto b_i^{-1}$ on Λ_A^0 to its saturation; it thus forms a torsor under $\text{hom}(\Lambda_A/\Lambda_A^0, \mathbb{C}^\times)$. Note that an *independent* A always has nonempty intersection.

For each layer W , we fix permanently an orientation

$$d_W \in \det \Lambda_W = \bigwedge^{\text{codim } W} \Lambda_W$$

with the convention $\mathbb{Z} \ni d_{\mathbf{T}} = 1$.

Set the notation $k_W := \text{codim } W$; then for $W \subset V$, the lattice Λ_W/Λ_V is a saturated sublattice of Λ/Λ_V . Define

$$\theta_{V,W} \in \bigwedge^{k_W - k_V} (\Lambda_W/\Lambda_V) \subset H^{k_W - k_V}(V; \mathbb{Z}) \quad (2)$$

by the requirement

$$\tilde{\theta}_{V,W} \wedge d_V = d_W \quad (3)$$

where here a tilde denotes a lift; the wedge is independent of the choice. We fix the normalization $\theta_{W,W} = 1$.

3 The cocharacter presentation

We now turn to constructing the various “chamber” decompositions via slicing the reified cocharacter lattices $N_{W,\mathbb{R}} := N_W \otimes \mathbb{R}$ along hyperplanes, where $N_W := \text{hom}(\Lambda_W, \mathbb{Z})$ is the cocharacter lattice of the quotient torus $\text{hom}(\Lambda_W, \mathbb{G}_m)$. In top degree (corresponding to the zero-dimensional layers), this directly extends the “spherical chains” parameterization [RX] for the arrangement of all central toric hyperplanes: our chambers will be strongly convex rational polyhedral cones, whose projection down to the unit sphere yields the translation to the language of polyhedral cells. (Note that in the universal setting of loc. cit., all such polyhedral cells decompose into simplicial ones, but this is rarely the case for a general finite arrangement.)

We have a tautological standard arrangement $\mathcal{A}[W]$ in $N_{W,\mathbb{R}}$ with hyperplanes defined by $\langle u_i, - \rangle = 0$ for $i \in S_W$. For $W \subset V$, we have also a restriction

$$\pi_{W,V} : N_{W,\mathbb{R}} \rightarrow N_{V,\mathbb{R}}.$$

Definition 3.1. Let $\mathcal{H}(\mathcal{A})$ be the graded R -module (which we will later endow with an algebra structure) with generators $f_{c,W}$ in degree k_W , as c ranges over the chambers $\text{Ch}(\mathcal{A}[W])$ of $\mathcal{A}[W]$, oriented by d_W . We impose the families of *degeneracy* relations

$$\psi_v f_{c,W} = 0, \quad \forall v \in \Lambda_W \quad (4)$$

and, for every $W \subset V$ and chamber c' of $\mathcal{A}[V]$, the *cylinder* relation

$$\theta_{V,W} f_{c',V} = \sum_{\substack{c \in \text{Ch}(\mathcal{A}[W]) \\ c \subset \pi_{W,V}^{-1}(c')}} f_{c,W}. \quad (5)$$

The rank-zero arrangement when $W = \mathbf{T}$ has one chamber; its generator will be the unit of $\mathcal{H}(\mathcal{A})$. Lifts of $\theta_{V,W}$ to R are immaterial by the degeneracy relation.

3.1 Geometric realization

As described in the introduction, we first realize our abstract presentation $\mathcal{H}(\mathcal{A})$ in the dual language of Borel–Moore homology rather than de Rham cohomology, since this will allow us to directly prove an *integral* isomorphism.

We define the layer quotient map

$$q_W : \mathbf{T} \rightarrow Q_W = \text{hom}(\Lambda_W, \mathbb{C}^\times), q_W(t)(v) := \lambda_W(v) \chi^v(t).$$

whose fiber over the identity is W . Write

$$\exp(a)(v) = \exp\langle v, a \rangle$$

for the exponential uniformization map $N_{W,\mathbb{R}} \rightarrow Q_W$. We associate to a chamber $c \in \text{Ch}(\mathcal{A}[W])$ the cycle

$$Z_{c,W} = q_W^{-1}(\exp c) \cap M,$$

a smooth real submanifold of dimension $2n - \text{codim } W$, which we orient according to the fixed choice d_W . The cycle $Z_{c,W}$ is closed in M , hence yields an integral Borel–Moore homology class $[Z_{c,W}] \in H_{2n-k_W}^{\text{BM}}(M)$. We have the corresponding Poincaré dual cohomology class $H^{k_W}(M) \ni z_{c,W} := \text{PD}[Z_{c,W}]$. Our main claim is:

Theorem 3.2. *The map*

$$\Phi : \mathcal{H}(\mathcal{A}) \longrightarrow H^\bullet(M; \mathbb{Z}), \quad \alpha f_{c,W} \mapsto \alpha|_M \smile z_{c,W} \quad (6)$$

is a well-defined isomorphism of graded modules, for any layer W , $\alpha \in R$, $c \in \text{Ch}(\mathcal{A}[W])$, which can be upgraded to an isomorphism of graded algebras (for the multiplicative structure specified in Section 5.4).

Well-definedness is relatively easy to show:

Proposition 3.3. *The map Φ is well-defined.*

Proof. It suffices to show the degeneracy and cylinder relations: for the degeneracy relation, observe that on $Z_{c,W}$, the argument of χ^v is constant for any $v \in \Lambda_W$. Thus, the restriction of ψ_v is zero.

For the cylinder relation, lift $\theta_{V,W}$ to a cohomology class $\psi_{v_1} \wedge \dots \wedge \psi_{v_d} \in R$ for a lift of an oriented basis of Λ_W/Λ_V ; this is Poincaré dual to the cycle cut out by the “phase” conditions

$$\lambda_W(v_i) \chi^{v_i}(z) \in \mathbb{R}_+$$

for $z \in \mathbf{T}$, with the orientation determined by d_W and d_V . The intersection of this locus with $Z_{c',V}$ is cut out by the equations $q_W(z) = \exp(x)$ as x ranges over the cylinder $x \in \pi_{W,V}^{-1}(c')$, which one may see decomposes into the various $Z_{c,W}$ for the subchambers $c \subset \pi_{W,V}^{-1}(c')$ as desired. $\square$

4 Proof of the presentation

In this section, we prove that Φ gives a presentation by reducing to local arguments, which amounts to reduction to the associated graded in the Leray spectral sequence.

4.1 Chambers and Orlik–Solomon classes

For a local calculation, we will find useful a relation between the chambers of a real hyperplane complement and the *classical* Orlik–Solomon model in the corresponding complexified hyperplane complement, which we recall now.

Let $\mathcal{B}$ be a central complexified real hyperplane arrangement in an oriented real affine space V of dimension q (from which we deduce all orientations on chambers). As in the toric case, we write for the *complexified* arrangement complement

$$M_{\mathcal{B}} = V_{\mathbb{C}} \setminus \bigcup_{H \in \mathcal{B}} H_{\mathbb{C}},$$

which is known by Brieskorn’s theorem to be integrally formal and torsion-free, with the Orlik–Solomon algebra of $\mathcal{B}$ giving a de Rham model for its cohomology respecting this integral structure.

Consider the free module $\mathbb{Z}\{\text{Ch}(\mathcal{B})\}$ on chambers, with generators written e_c for $c \in \text{Ch}(\mathcal{B})$; we wish to impose certain “lineal” relations: for a flat X (i.e. an intersection of some subset of hyperplanes), write

$$\mathcal{B}_X = \{H/X : H \supset X\}$$

in V/X for the quotiented arrangement; for each chamber D' of this localization, write

$$r_{X,D'} = \sum_{c \subset \pi_X^{-1}(D')} e_c$$

For nontrivial X , these are precisely “lineal” unions of chambers, which contain some entire subspace. We define the relations module $\mathcal{R}$ to be the subspace of $\mathbb{Z}\{\text{Ch}(\mathcal{B})\}$ spanned by such lineal combinations of chambers. Note that the relations for higher-dimensional X clearly follow from those for lower-dimensional, so if $\mathcal{B}$ is essential, then it suffices to take X one-dimensional.

Proposition 4.1. *The map $e_c \mapsto [c]$ induces an isomorphism*

$$\mathbb{Z}\{\text{Ch}(\mathcal{B})\}/\mathcal{R} \xrightarrow{\sim} H_q^{\text{BM}}(M_{\mathcal{B}}; \mathbb{Z}) \cong H^q(M_{\mathcal{B}}; \mathbb{Z}). \quad (7)$$

Proof. First, observe that the relations certainly do hold in the target, so the map is well-defined: indeed, note that the image of a lineal combination of chambers actually has boundary lying only on the hyperplanes containing X , a non-essential subarrangement of lower rank which thus has vanishing degree- q cohomology.

To prove that it is an isomorphism, we construct an inverse: the Orlik–Solomon model furnishes an integral presentation of the right-hand side in terms of homogeneous antisymmetric symbols $[\ell_1, \dots, \ell_q]$ for real linear functionals $\ell_1, \dots, \ell_q$ cutting out hyperplanes of $\mathcal{B}$, subject to cyclic circuit relations among dependent $(q+1)$ -tuples. Here, the de Rham realization of $[\ell_1, \dots, \ell_q]$ is (up to the q th power of the Tate period) $d \log \ell_1 \wedge \dots \wedge d \log \ell_q$.

Given a linear functional ℓ , write C_{ℓ} for the dual cone in V where it is positive. We then send the Orlik–Solomon monomial $[\ell_1, \dots, \ell_q]$ to

$$\text{sgn} \begin{pmatrix} \ell_1 & \dots & \ell_q \end{pmatrix} \sum_{c \subset \bigcap_i C_{\ell_i}} [c] \in \mathbb{Z}\{\text{Ch}(\mathcal{B})\}/\mathcal{R}$$

This is well-defined: antisymmetry is built in, scaling linear functionals by a positive constant does not affect the dual cones, scaling by -1 changes the result by a lineal combination of cones, and the circuit relations map to subdivision relations of polyhedral cones.

To check the two composites are the identity, we claim that a Poincaré dual to $[c]$ can be constructed by subdividing c into a set of simplicial cones Σ_c and then taking the sum

$$\sum_{\tau \in \Sigma_c} [\tau_1, \dots, \tau_q]$$

where $\tau_1, \dots, \tau_q$ are the linear forms cutting out the hyperplanes bounding τ , with orientation coming from the orientation of c . Indeed, by functoriality of the Orlik–Solomon model and the injectivity of its complement pullback maps along inclusions of arrangements, we may pass without losing information to a larger real arrangement in which these simplicial cones *are* generated; i.e. we may assume c itself is simplicial without loss of generality. Using functoriality of the Orlik–Solomon model under $\mathrm{GL}_q(\mathbb{R})$ to move the cone around, we are reduced to the case of the standard chamber bounded by $e_1, \dots, e_q$, in an arrangement containing the standard coordinate hyperplanes, being dual to $(2\pi i)^{-q} d \log z_1 \wedge \dots \wedge d \log z_q$; this then reduces to the one-dimensional case, where one can compute directly that $(2\pi i)^{-1} d \log z$ is dual to $\mathbb{R}_{\geq 0}$.

Accepting the claim, both composites can then be computed to be the identity via inclusion-exclusion. $\square$

We remark this proposition is precisely the “finite arrangement” version of the fact that the quotient of a module of indicator functions of rational polyhedral cones by all lower-dimensional and lineal cones is the top-degree part of an Orlik–Solomon complex, already written in [X1] and [X2] in the case of the pro-arrangement of *all* rational hyperplanes.

We will apply our proposition to the real arrangements $M(\mathcal{A}[W])$ in the reified cocharacter lattice $N_{W, \mathbb{R}}$; we denote the corresponding Orlik–Solomon cohomology module by

$$A_W = H^{k_W}(M(\mathcal{A}[W]); \mathbb{Z}).$$

4.2 The Leray spectral sequence

Write $j : M \hookrightarrow \mathbf{T}$ and $i_W : W \hookrightarrow \mathbf{T}$ for the inclusions.

Proposition 4.2. *We have the generalized Orlik–Solomon decomposition*

$$R^q j_* \mathbb{Z} \cong \bigoplus_{k_W=q} (i_W)_* \underline{A}_W. \quad (8)$$

Proof. Hodge-theoretic arguments to establish this rationally are well-known in the literature to establish the corresponding Leray spectral sequence of mixed Hodge structures; cf. [Du, Theorem 3.1].

To obtain an integral version, we note that in a small neighborhood of a point p in the relatively open stratum of a layer W , the neighborhood looks like the complement of the hypertori H_i , $i \in S_W$, in a small normal slice to W at p , times a bunch of contractible directions; the former looks like (and can be identified via the exponential map with) the complex hyperplane arrangement $\mathcal{A}[W]$, which canonically identifies the stalk with A_W .

Along the relatively open stratum, these glue in the obvious way; if p is instead inside a stratum W strictly contained in W' , then one may compute that the cospecialization map $A_W^q \rightarrow A_{W'}^q$, exactly is the identity on Orlik–Solomon generators $[\ell_1, \dots, \ell_q]$ for which all (exponentiated) linear forms cut out hyperplanes also part of the arrangement $\mathcal{A}[W']$, and kills the others (as they can be

checked simply on a local uniformizer f , where $d \log f$ becomes exact once one shrinks a small ball on the larger stratum).

Thus, the stalkwise restriction maps assemble to (8). $\square$

We consequently obtain a multiplicative Leray spectral sequence

$$E_2^{p,q} = \bigoplus_{k_W=q} H^p(W; \mathbb{Z}) \otimes A_W \implies H^{p+q}(M; \mathbb{Z}) \quad (9)$$

of R -modules (acting through restriction to the layers W). The edge map in total degree q is the direct sum of local restrictions, which may be defined geometrically just as in the preceding proof: given a layer W' with $k_{W'} = q$, and a point p in the relatively open stratum of W' , the edge map $H^q(M; \mathbb{Z}) \rightarrow A_{W'}$ can be computed geometrically as the restriction

$$\rho_{W'} : H^q(M; \mathbb{Z}) \rightarrow H^q(B_p^\perp \cap M; \mathbb{Z}). \quad (10)$$

where $B_p^\perp$ is a small transverse ball to the layer at p , whose intersection with M may be identified via the exponential with the germ of the local hyperplane arrangement.

Our cospecialization argument implies:

Lemma 4.3. *For $k_W = k_{W'} = q$, the integral geometric classes satisfy*

$$\rho_{W'}(z_{c,W}) = \begin{cases} \text{PD}[c] & W' = W, \\ 0 & W' \neq W. \end{cases} \quad (11)$$

Remark 4.4. This material is also implicitly used in [CDDMP, Definition 5.13] and what follows, but its justification is omitted; only the analogous rational-coefficients (originally given Hodge-theoretic arguments) results are cited. We provide proofs here for the sake of completeness.

4.3 Degeneration and the proof of the integral presentation

By Proposition 4.1 and Lemma 4.3, the classes $z_{c,W}$ lift generators of every $E_2^{0,q}$, hence survive to the infinity page. Noting that the differentials are R -module morphisms, and that R surjects onto $H^*(W; \mathbb{Z})$, we deduce that the whole E_2 -page is generated as an R -module by our cycles. The differentials vanish on the row $q = 0$ for degree reasons and on the chosen lifts since they are permanent; by a similar argument, we conclude that all subsequent differentials vanish and the spectral sequence collapses.

Since the E_2 terms are all free abelian, we deduce that $H^\bullet(M; \mathbb{Z})$ is free abelian in each degree. In total degree d , we reindex the Leray filtration so that it has graded pieces

$$\text{gr}_k H^d(M; \mathbb{Z}) = \bigoplus_{k_W=k} H^{d-k}(W; \mathbb{Z}) \otimes A_W. \quad (12)$$

We filter $\mathcal{H}$ by degree of chamber generators:

$$F_k \mathcal{H} = \sum_{k_W \leq k} R f_{c,W}. \quad (13)$$

In the graded degree- k part gr_k , a cylinder relation from $V \supsetneq W$ has vanishing left-hand side. The degeneracy relations mean we may think of the R -algebra structure as factoring through $H^*(W; \mathbb{Z})$,

and these graded cylinder relations then precisely reduce to those recorded in Proposition 4.1. We thus have produced a surjection

$$\bigoplus_{k_W=k} H^{d-k}(W; \mathbb{Z}) \otimes A_W \rightarrow \mathrm{gr}_k \mathcal{H}^d. \quad (14)$$

To prove it is also injective, we note our map Φ is filtered: $z_{c,W}$ has total degree k_W , and multiplying by a degree- p class sends it to $F_{k_W} H^{p+k_W}$. If we compose (14) with $\mathrm{gr}_k \Phi$ and (12), then our previous residue calculation identifies this with the identity on each layer summand. Since the first map is surjective, both maps are isomorphisms. Since passing to graded pieces is reflective, we conclude Φ is an isomorphism.

5 The character formulation and de Rham realization

We give now the analogue/generalization of the map Ψ (from [X1], [RX]) realized in the “magical ring” of log-pole differential forms realizing the formality isomorphism for $\mathbf{T} - H$ (cf. [BD]); this recovers a presentation equivalent to that given in [CDDMP]. This map coincides with our cycle-defined Φ under the Betti–de Rham isomorphism.

As in [RX], the “dual conical” formulation to our conical chambers is useful here, since the corresponding forms can be realized as Shintani-style generating series of these cones.

5.1 Admissible cones and their decorations

We define an *admissible cone* over W to be a closed cone generated by a subset of $\{\pm u_i : i \in S_W\}$. If τ is such a cone, there is a unique layer $W(\tau) \supset W$ with

$$\Lambda_{W(\tau)} = \Lambda \cap \mathrm{span}_{\mathbb{R}} \tau, \quad \lambda_{W(\tau)} = \lambda_W|_{\Lambda_{W(\tau)}}. \quad (15)$$

Indeed, choose a generating subset for τ ; then $W(\tau)$ is the component of the corresponding hypertorus intersection containing W . Equivalently, it is the connected coset cut out by (15) (showing independence of the generating set).

A decorated cone is a pair (σ, W) with σ admissible over W and real span $\Lambda_{W, \mathbb{R}}$. The zero cone is decorated by $\mathbf{T}$. Note that passing to a face changes the decoration by (15).

5.2 Conical generating functions

For a pointed strongly convex rational polyhedral cone $\sigma \subset \Lambda_{W, \mathbb{R}}$, of dimension r , define

$$g_W(\sigma) = (-1)^r \sum_{v \in \sigma^\circ \cap \Lambda_W} \lambda_W(v) \chi^v, \quad (16)$$

where here $\circ$ denotes relative interior; this converges (on some domain, by strong convexity) to a rational function. For not necessarily-strongly convex cones, their generating functions can be defined consistently by decomposing into strongly convex ones; cones with lineality satisfy $g_W(\sigma) = 0$. This defines a valuation on such cones, in that it factors through the module of their indicator functions modulo lineal cones. (This is all classical material, used e.g. in Brion et al.’s work on counting lattice points, or Dasgupta et al.’s work on Shintani cocycles.)

Note that the dependence on W is weak; restriction to a smaller layer for which the cone remains admissible does not change the series:

$$g_W(\tau) = g_{W(\tau)}(\tau). \quad (17)$$

As in [X1] and [RX] (and in [CDDMP], in the language of coverings over which arrangements “separate”), we may also write these generating series as pushforwards: suppose $\sigma_A = \text{cone}(u_i : i \in A)$ for an independent set of primitive character vectors $A \subset S_W$ of size k . Let $v_1, \dots, v_k$ be primitive cocharacters bounding the dual cone,² $\sigma_A^\vee$, ordered compatibly with the chosen orientation d_W , and let

$$\varphi_A : (\mathbb{G}_m)^k \longrightarrow Q_W = \text{Hom}(\Lambda_W, \mathbb{G}_m), (z_1, \dots, z_k) \mapsto \prod_{j=1}^k v_j(z_j).$$

Then

$$g_W(\sigma_A) d_W = q_W^*(\varphi_A)_* \left(\bigwedge_{j=1}^k \frac{1}{2\pi i} \frac{dz_j}{z_j - 1} \right). \quad (18)$$

As is common in the Shintani literature, the pushforward may also be instead written by summing cosets over the “unit box” $P_A^\circ = \{\sum_{i \in A} z_i u_i : 0 < z_i \leq 1\}$, as

$$g_W(\sigma_A) = (-1)^k \frac{\sum_{v \in P_A^\circ \cap \Lambda_W} \lambda_W(v) \chi^v}{\prod_{i \in A} (1 - b_i^{-1} \chi^{u_i})}. \quad (19)$$

Definition 5.1. Let $\mathcal{V}(\mathcal{A})$ be the graded R -module generated by $e_{\sigma, W}$, in degree k_W , for all decorated cones, subject to:

1. (lineality) $e_{\sigma, W} = 0$ if σ is lineal (i.e. contains a nonzero subspace);
2. (degeneracy) $\psi_v e_{\sigma, W} = 0$ for $v \in \Lambda_W$;
3. (subdivision) for every W and every indicator-function relation $\sum_j a_j \mathbf{1}_{\tau_j} = 0$ among admissible cones in $\Lambda_{W, \mathbb{R}}$

$$\sum_j a_j \theta_{W(\tau_j), W} e_{\tau_j, W(\tau_j)} = 0. \quad (20)$$

Now, let K_W be the group generated by indicators of closed cones in the *cocharacter* lattice N_W cut out by any subset of the linear functionals $\pm \langle u_i, - \rangle$, and K_W^0 be the subgroup of functions vanishing on the complement of these hyperplanes, so that

$$K_W / K_W^0 = \mathbb{Z}^{\text{Ch}(\mathcal{A}[W])}, \quad (21)$$

with basis the chamber indicator functions.

Let P_W be the group generated by indicators of *all* admissible character cones over W , including those with smaller span, and let L_W be the subgroup generated by its lineal cones. Then conical duality for the reified character/cocharacter lattices $\Lambda_{W, \mathbb{R}}$ and $N_{W, \mathbb{R}}$ gives

$$D : P_W / L_W \xrightarrow{\sim} K_W / K_W^0, \quad D(\mathbf{1}_\sigma) = \mathbf{1}_{\sigma^\vee}. \quad (22)$$

Indeed, the indicators of lower-dimensional arrangement faces span K_W^0 , and their duals are exactly the lineal cones.

Theorem 5.2. *Duality induces an isomorphism $D : \mathcal{H}(\mathcal{A}) \cong \mathcal{V}(\mathcal{A})$, characterized by*

$$f_{c, W} = e_{c^\vee, W}. \quad (23)$$

²In the cocharacter formulation, this corresponds to the Poincaré dual of a sum of chambers giving a simplicial cone.

Proof. For fixed W , apply (22) to all its cone identities. The dual in $N_{W,\mathbb{R}}$ of a cone τ spanning $\Lambda_{V,\mathbb{R}}$ is the inverse image of its dual in $N_{V,\mathbb{R}}$. Consequently, the identification of its smaller-span symbol with the same indicator in P_W , after multiplication by $\theta_{V,W}$, becomes the cylinder relation. It suffices to impose this for chamber indicators since they form the basis in (21). Conversely, expanding cylinders as sums of chambers recovers every such smaller-span identification, while (22) recovers all cone subdivision relations and the lineality relations. The degeneracy relations are the same on both sides. $\square$

5.3 Differential form representatives

Let $\Omega(\mathcal{A})$ be the “magic ring” of differential forms with log poles on M .

Theorem 5.3. *The previously-defined valuation yields a realization*

$$\Psi : \mathcal{V}(\mathcal{A}) \rightarrow \Omega(\mathcal{A}), \quad \Psi(e_{\sigma,W}) := g_W(\sigma)d_W, \quad \Psi(f_{c,W}) := g_W(c^\vee)d_W, \quad (24)$$

noting that indeed $g_W(\sigma)d_W$ is closed, and has all poles lying on hypertori of $\mathcal{A}$.

Proof. The degeneracy relation follows from $\psi_v \wedge d_W = 0$ for $v \in \Lambda_W$. For (20), equations (3) and (17) turn its image into

$$\sum_j a_j g_W(\tau_j)d_W = 0,$$

which is exactly the subdivision relation. Lineality is killed by definition. $\square$

Proposition 5.4. *For every symbol $h \in \mathcal{H}(\mathcal{A})$, the de Rham class of $\Psi(D(h))$ is the complexification of $\Phi(h)$. In particular these forms represent Betti-integral classes, and Ψ identifies $\mathcal{H}$ with its algebra of closed-form representatives.*

Proof. In rank 1, as we previously computed, the class of $(1, \infty) \subset \mathbb{G}_m \setminus \{1\}$ is Poincaré dual to $\omega := (2\pi i)^{-1} d \log(1 - z)$. Similarly, one computes that the other chamber yields the cycle $(0, 1)$, dual to $\psi - \omega$.

By the subdivision relations, it suffices to consider an independent symbol, together with its products with various ψ 's. Let $c = \sigma_A^\vee$, and use the isogeny φ_A and translated quotient q_W above. The restriction of φ_A induces an orientation-preserving diffeomorphism

$$(\mathbb{R}_{>0})^k \longrightarrow \exp(c).$$

Here the source denotes the region $z_j > 1$ for all j , identified with the positive orthant via logarithm. The form

$$\bigwedge_{j=1}^k \frac{1}{2\pi i} \frac{dz_j}{z_j - 1}$$

represents the Poincaré dual of this region in $(\mathbb{C}^\times \setminus \{1\})^k$, by taking products in the rank-1 computation.

Poincaré duality is compatible with proper pushforward and smooth pullback. Applying $(\varphi_A)_*$ and then q_W^* , and restricting to the arrangement complement, thus identifies the de Rham class of $g_W(\sigma_A)d_W$ with $z_{c,W}$. Extending by the R -action, and extending to arbitrary h by the subdivision relations, the result follows. $\square$

5.4 Multiplicative structure

The module presentation admits the following explicit product, which can be checked via the differential forms (and then also captures the whole integral multiplicative structure of cohomology via the preceding results). Write $k = k_W$, $k' = k_{W'}$. If $\Lambda_{W,\mathbb{Q}} \cap \Lambda_{W',\mathbb{Q}} \neq 0$, set $e_{\sigma,W}e_{\tau,W'} = 0$.

Otherwise, for every component L of $W \cap W'$, let

$$d_W \wedge d_{W'} = \epsilon(W, W', L) d d_L, \quad d = [\Lambda_L : \Lambda_W + \Lambda_{W'}], \quad \epsilon(W, W', L) \in \{\pm 1\}.$$

Then

$$e_{\sigma,W}e_{\tau,W'} = \sum_{L \in \pi_0(W \cap W')} \epsilon(W, W', L) e_{\sigma+\tau,L}. \quad (25)$$

We extend by the usual graded sign rule:

$$(\alpha e_{\sigma,W})(\beta e_{\tau,W'}) = (-1)^{k \deg \beta} \alpha \beta (e_{\sigma,W}e_{\tau,W'}).$$

As mentioned previously, the unit is $e_{\{0\},\mathbf{T}}$.

To see why (25) holds, note that (when spans are independent) the two twists give one character λ_0 on $L_0 = \Lambda_W \oplus \Lambda_{W'}$. There are d extensions to Λ_L , indexed by the components of $W \cap W'$. Character orthogonality then tells us

$$\sum_{\lambda|_{L_0}=\lambda_0} \lambda(v) = \begin{cases} d \lambda_0(v), & v \in L_0, \\ 0, & v \notin L_0. \end{cases}$$

Since addition gives a bijection $(\sigma^\circ \cap \Lambda_W) \times (\tau^\circ \cap \Lambda_{W'}) \rightarrow (\sigma + \tau)^\circ \cap L_0$, it follows that

$$\sum_L g_L(\sigma + \tau) = d g_W(\sigma) g_{W'}(\tau). \quad (26)$$

The multiplicity is the same factor occurring in $d_W \wedge d_{W'}$; (25) follows.

Using the duality identification $\mathcal{H}(\mathcal{A}) \cong \mathcal{V}(\mathcal{A})$, we may describe the same product structure on the cocharacter side by pulling back chambers c and c' to $N_{L,\mathbb{R}}$, intersecting the two cylinders, and summing over all chambers of $\mathcal{A}[L]$ contained in that intersection (with the same orientation sign and sum over L). (If the spans are not independent, the product is again zero.)

References

- [BD] Francis Brown and Clément Dupont. “Positive geometries and canonical forms via mixed Hodge theory”. In: *Communications in Mathematical Physics* 406.11, 267 (2025).
- [CDDMP] Filippo Callegaro et al. “Orlik–Solomon-type presentations for the cohomology algebra of toric arrangements”. In: *Transactions of the American Mathematical Society* 373.3 (2020), pp. 1909–1940.
- [CDDMP2] Filippo Callegaro et al. “Corrigendum to “Orlik–Solomon-type presentations for the cohomology algebra of toric arrangements””. In: *Transactions of the American Mathematical Society* 374.5 (2021), pp. 3779–3781.
- [CRR] Steven Charlton, Danylo Radchenko, and Daniil Rudenko. “Multiple polylogarithms and the Steinberg module”. Preprint. 2025.
- [DP] Corrado De Concini and Claudio Procesi. “On the geometry of toric arrangements”. In: *Transformation Groups* 10.3-4 (2005), pp. 387–422.

- [Du] Clément Dupont. “Purity, formality, and arrangement complements”. In: *International Mathematics Research Notices* 2016.13 (2016), pp. 4132–4144.
- [OS] Peter Orlik and Louis Solomon. “Combinatorics and topology of complements of hyperplanes”. In: *Inventiones Mathematicae* 56.2 (1980), pp. 167–189.
- [RX] Peter Xu and Marti Roset Julia. “Rigid analytic Eisenstein cocycles for $\mathrm{SL}_n(\mathbb{Z}[1/p])$ ”. Preprint. 2026.
- [X1] Peter Xu. “Symbols for toric Eisenstein cocycles and arithmetic applications”. Preprint. 2025.
- [X2] Peter Xu. “A note on polyhedral cones and toric polylogarithms”. Preprint. 2026.